Lab 7: Fuzzing for Input Validation Bugs

Name	Sripada Krishna Sameer Aditya
SRN	PES1UG24AM285
Assigned repo	SETAPESU26/L6_L7_46_photo_print_order (Lab 7)
Tool	Hypothesis 6.170.15, pytest 9.2.0, Python 3.13

The handout's `processor.py` / `test_processor.py` correspond to `photo_print_order.py` / `test_photo_print_order.py` in this repo.

Bugs found (original code)

Function	Input	Result
`parse_order`	None / wrong type	AttributeError: 'NoneType' has no attribute 'split'
`parse_order`	"4x6" (no colon), "a:b:c"	ValueError on tuple unpacking
`parse_order`	"4x6:abc"	ValueError from int()
`parse_order`	":5"	silently returns ('', 5): empty size accepted
`parse_order`	"4x6:-1"	silently returns ('4X6', -1): negative copies accepted
`file_size_average`	"", "4,,8", "a,b", None	ValueError / AttributeError
`file_size_average`	"4,-8"	returns -2: negative file size accepted
`payable_amount`	"450", "450|", "1|2|3", None	ValueError / AttributeError
`payable_amount`	"nan|1", "inf|1"	returns nan / inf instead of an amount
`payable_amount`	"5|-100"	returns 105.9: a negative discount raises the price
`payable_amount`	"1.7e308|0"	finite input, result overflows to inf

Fix summary

Each function now returns `None` for invalid input and never raises: type check, exact delimiter count, `try/except ValueError` around `int()`/`float()`, rejection of negative and non-finite (nan/inf) numbers, and an overflow check on the total. `main()` prints a friendly message when a result is `None`. Result: 61 of 69 tests failed before; all 69 pass after.

Step 2: Run the buggy application

`python photo_print_order.py`: normal input works, but an empty list or missing discount crashes with a traceback.

Step 2 - buggy photo_print_order.py (Lab 7)

```
$ python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0


$ python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12):
Traceback (most recent call last):
  File "/home/claude/lab7/work/photo_print_order.py", line 31, in <module>
    main()
    ~~~~^^
  File "/home/claude/lab7/work/photo_print_order.py", line 22, in main
    result = file_size_average(user_input)
  File "/home/claude/lab7/work/photo_print_order.py", line 6, in file_size_average
    values = [int(r) for r in sizes_csv.split(',')]
              ~~~^^^
ValueError: invalid literal for int() with base 10: ''

$ python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|
Traceback (most recent call last):
  File "/home/claude/lab7/work/photo_print_order.py", line 31, in <module>
    main()
    ~~~~^^
  File "/home/claude/lab7/work/photo_print_order.py", line 27, in main
    result = payable_amount(user_input)
  File "/home/claude/lab7/work/photo_print_order.py", line 11, in payable_amount
    total = float(amount) * 1.18 - float(discount)
                                   ~~~~~^^^^^^^^^^
ValueError: could not convert string to float: ''
```

Step 4: pytest against the buggy code (failing)

Hypothesis finds minimal failing inputs by itself (`':'`, `''`, `None`, ...).

```
Step 4 - pytest vs BUGGY photo_print_order.py: Hypothesis falsifying examples

$ pytest test_photo_print_order.py -k "test_parse_order_no_crash or test_file_size_average_no_crash or test_payable_a
============================================= test session starts ==============================================
platform linux -- Python 3.13.17, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/claude/lab7/work
plugins: platformdirs-4.13.0, anyio-4.15.1, hypothesis-6.170.15
collected 69 items / 66 deselected / 3 selected

test_photo_print_order.py FFF                                                                          [100%]

=================================================== FAILURES ===================================================
_____________________________________ test_parse_order_no_crash ______________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/claude/lab7/work/test_photo_print_order.py", line 38, in test_parse_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 40, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 3, in parse_order
    |     return size.strip().upper(), int(copies)
    |                               ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 42, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e!r}"
    | AssertionError: parse_order crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 40, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 2, in parse_order
    |     size, copies = text.split(':')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 42, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e!r}"
    | AssertionError: parse_order crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 40, in test_parse_order_no_crash
```

```
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 40, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 2, in parse_order
    |     size, copies = text.split(':')
    |                    ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 42, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e!r}"
    | AssertionError: parse_order crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________ test_file_size_average_no_crash ______________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/claude/lab7/work/test_photo_print_order.py", line 46, in test_file_size_average_no_crash
  |     def test_file_size_average_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 48, in test_file_size_average_no_crash
    |     file_size_average(s)
    |     ~~~~~~~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 6, in file_size_average
    |     values = [int(r) for r in sizes_csv.split(',')]
    |               ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 50, in test_file_size_average_no_crash
    |     assert False, f"file_size_average crashed with: {e!r}"
    | AssertionError: file_size_average crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_file_size_average_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 48, in test_file_size_average_no_crash
    |     file_size_average(s)
    |     ~~~~~~~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 6, in file_size_average
    |     values = [int(r) for r in sizes_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 50, in test_file_size_average_no_crash
    |     assert False, f"file_size_average crashed with: {e!r}"
    | AssertionError: file_size_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
```

```
    | assert False
    | Failing test case: test_file_size_average_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________ test_payable_amount_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/claude/lab7/work/test_photo_print_order.py", line 54, in test_payable_amount_no_crash
  |     def test_payable_amount_no_crash(s):
  |                  ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 56, in test_payable_amount_no_crash
    |     payable_amount(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 11, in payable_amount
    |     total = float(amount) * 1.18 - float(discount)
    |             ~~~~~^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 58, in test_payable_amount_no_crash
    |     assert False, f"payable_amount crashed with: {e!r}"
    | AssertionError: payable_amount crashed with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_payable_amount_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 56, in test_payable_amount_no_crash
    |     payable_amount(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 10, in payable_amount
    |     amount, discount = text.split('|')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 58, in test_payable_amount_no_crash
    |     assert False, f"payable_amount crashed with: {e!r}"
    | AssertionError: payable_amount crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_payable_amount_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 56, in test_payable_amount_no_crash
    |     payable_amount(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "/home/claude/lab7/work/photo_print_order.py", line 10, in payable_amount
    |     amount, discount = text.split('|')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/claude/lab7/work/test_photo_print_order.py", line 58, in test_payable_amount_no_crash
    |     assert False, f"payable_amount crashed with: {e!r}"
    | AssertionError: payable_amount crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_payable_amount_no_crash(
    |     s=None,
    | )
    +------------------------------------
========================================== short test summary info ==========================================
FAILED test_photo_print_order.py::test_parse_order_no_crash - ExceptionGroup: Hypothesis found 3 distinct f...
FAILED test_photo_print_order.py::test_file_size_average_no_crash - ExceptionGroup: Hypothesis found 2 dist...
FAILED test_photo_print_order.py::test_payable_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinc...
====================================== 3 failed, 66 deselected in 0.67s ======================================
```

Step 4 (cont.): full-suite summary, 61 failed, 8 passed

pytest test_photo_print_order.py

```
Step 4 - pytest vs BUGGY photo_print_order.py: full-suite summary

$ pytest test_photo_print_order.py -q --tb=no -rf
FFFFFFFFFFFFFFFFFFFFFFFF..FFFFFFFFFF..FFFFFFFFFF....FFFFFFFFFFFFFFFF                                    [100%]
========================================= short test summary info ==========================================
FAILED test_photo_print_order.py::test_parse_order_no_crash - ExceptionGroup: Hypothesis found 3 distinct f...
FAILED test_photo_print_order.py::test_file_size_average_no_crash - ExceptionGroup: Hypothesis found 2 dist...
FAILED test_photo_print_order.py::test_payable_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinc...
FAILED test_photo_print_order.py::test_parse_order_any_type_no_crash - ExceptionGroup: Hypothesis found 4 d...
FAILED test_photo_print_order.py::test_file_size_average_any_type_no_crash - ExceptionGroup: Hypothesis fou...
FAILED test_photo_print_order.py::test_payable_amount_any_type_no_crash - ExceptionGroup: Hypothesis found ...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[None-parse_order] - AttributeError: 'N...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[None-file_size_average] - AttributeErr...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[None-payable_amount] - AttributeError:...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[5-parse_order] - AttributeError: 'int'...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[5-file_size_average] - AttributeError:...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[5-payable_amount] - AttributeError: 'i...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[4.5-parse_order] - AttributeError: 'fl...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[4.5-file_size_average] - AttributeErro...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[4.5-payable_amount] - AttributeError: ...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[4x6:25-parse_order] - TypeError: a byt...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[4x6:25-file_size_average] - TypeError:...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[4x6:25-payable_amount] - TypeError: a ...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[bad4-parse_order] - AttributeError: 'l...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[bad4-file_size_average] - AttributeErr...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[bad4-payable_amount] - AttributeError:...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[bad5-parse_order] - AttributeError: 'd...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[bad5-file_size_average] - AttributeErr...
FAILED test_photo_print_order.py::test_non_string_input_returns_none[bad5-payable_amount] - AttributeError:...
FAILED test_photo_print_order.py::test_parse_order_without_colon_is_invalid - ValueError: not enough values...
FAILED test_photo_print_order.py::test_parse_order_too_many_colons_is_invalid - ValueError: too many values...
FAILED test_photo_print_order.py::test_parse_order_negative_copies_is_invalid - AssertionError: assert ('0X...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[:5] - AssertionError: assert ('', 5) is...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[  :5] - AssertionError: assert ('', 5) ...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:] - ValueError: invalid literal for...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:abc] - ValueError: invalid literal ...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:2.5] - ValueError: invalid literal ...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:-1] - AssertionError: assert ('4X6'...
FAILED test_photo_print_order.py::test_parse_order_known_bad_inputs[] - ValueError: not enough values to un...
FAILED test_photo_print_order.py::test_file_size_average_negative_size_is_invalid - AssertionError: assert ...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[] - ValueError: invalid literal f...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[,] - ValueError: invalid literal ...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[4,,8] - ValueError: invalid liter...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[4,8,] - ValueError: invalid liter...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[,4] - ValueError: invalid literal...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[a,b] - ValueError: invalid litera...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[4;8] - ValueError: invalid litera...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[1.5,2] - ValueError: invalid lite...
FAILED test_photo_print_order.py::test_file_size_average_known_bad_inputs[   ] - ValueError: invalid litera...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[] - ValueError: not enough values to...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[450] - ValueError: not enough values...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[450|] - ValueError: could not conver...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[|50] - ValueError: could not convert...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[|] - ValueError: could not convert s...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[a|b] - ValueError: could not convert...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[1|2|3] - ValueError: too many values...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[nan|1] - AssertionError: assert nan ...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[1|nan] - AssertionError: assert nan ...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[inf|1] - AssertionError: assert inf ...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[1|inf] - AssertionError: assert 0 is...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[-inf|1] - AssertionError: assert 0 i...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[1e999|0] - AssertionError: assert in...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[-5|1] - AssertionError: assert 0 is ...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[5|-1] - AssertionError: assert 6.899...
FAILED test_photo_print_order.py::test_payable_amount_known_bad_inputs[1.7e308|0] - AssertionError: assert ...
FAILED test_photo_print_order.py::test_payable_amount_needs_exactly_one_pipe - ValueError: not enough value...
61 failed, 8 passed in 1.20s
```

Step 5: Run the fixed application

Same valid output as before; invalid input is now rejected with a clear message instead of a crash.

Step 5 - FIXED photo_print_order.py

```
$ python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0


$ python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6
Invalid input. Expected size:copies, e.g. 4x6:25

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12):
Invalid input. Expected whole-number sizes, e.g. 4,8,12

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|
Invalid input. Expected amount|discount, e.g. 450|50

$ python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:abc
Invalid input. Expected size:copies, e.g. 4x6:25

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,-8
Invalid input. Expected whole-number sizes, e.g. 4,8,12

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): nan|5
Invalid input. Expected amount|discount, e.g. 450|50
```

Step 6: pytest against the fixed code (passing)

All 69 tests pass.

```
Step 6 - pytest vs FIXED photo_print_order.py

$ pytest test_photo_print_order.py -v
====================================== test session starts ======================================
platform linux -- Python 3.13.17, pytest-9.2.0, pluggy-1.7.0 -- /usr/local/bin/python3
hypothesis profile 'default'
rootdir: /home/claude/lab7/work
plugins: platformdirs-4.13.0, anyio-4.15.1, hypothesis-6.170.15
collecting ... collected 69 items

test_photo_print_order.py::test_parse_order_no_crash PASSED                                  [  1%]
test_photo_print_order.py::test_file_size_average_no_crash PASSED                            [  2%]
test_photo_print_order.py::test_payable_amount_no_crash PASSED                               [  4%]
test_photo_print_order.py::test_parse_order_any_type_no_crash PASSED                         [  5%]
test_photo_print_order.py::test_file_size_average_any_type_no_crash PASSED                   [  7%]
test_photo_print_order.py::test_payable_amount_any_type_no_crash PASSED                      [  8%]
test_photo_print_order.py::test_non_string_input_returns_none[None-parse_order] PASSED       [ 10%]
test_photo_print_order.py::test_non_string_input_returns_none[None-file_size_average] PASSED [ 11%]
test_photo_print_order.py::test_non_string_input_returns_none[None-payable_amount] PASSED    [ 13%]
test_photo_print_order.py::test_non_string_input_returns_none[5-parse_order] PASSED          [ 14%]
test_photo_print_order.py::test_non_string_input_returns_none[5-file_size_average] PASSED    [ 15%]
test_photo_print_order.py::test_non_string_input_returns_none[5-payable_amount] PASSED       [ 17%]
test_photo_print_order.py::test_non_string_input_returns_none[4.5-parse_order] PASSED        [ 18%]
test_photo_print_order.py::test_non_string_input_returns_none[4.5-file_size_average] PASSED  [ 20%]
test_photo_print_order.py::test_non_string_input_returns_none[4.5-payable_amount] PASSED     [ 21%]
test_photo_print_order.py::test_non_string_input_returns_none[4x6:25-parse_order] PASSED     [ 23%]
test_photo_print_order.py::test_non_string_input_returns_none[4x6:25-file_size_average] PASSED [ 24%]
test_photo_print_order.py::test_non_string_input_returns_none[4x6:25-payable_amount] PASSED  [ 26%]
test_photo_print_order.py::test_non_string_input_returns_none[bad4-parse_order] PASSED       [ 27%]
test_photo_print_order.py::test_non_string_input_returns_none[bad4-file_size_average] PASSED [ 28%]
test_photo_print_order.py::test_non_string_input_returns_none[bad4-payable_amount] PASSED    [ 30%]
test_photo_print_order.py::test_non_string_input_returns_none[bad5-parse_order] PASSED       [ 31%]
test_photo_print_order.py::test_non_string_input_returns_none[bad5-file_size_average] PASSED [ 33%]
test_photo_print_order.py::test_non_string_input_returns_none[bad5-payable_amount] PASSED    [ 34%]
test_photo_print_order.py::test_parse_order_valid_round_trip PASSED                          [ 36%]
test_photo_print_order.py::test_parse_order_ignores_surrounding_whitespace PASSED            [ 37%]
test_photo_print_order.py::test_parse_order_without_colon_is_invalid PASSED                  [ 39%]
test_photo_print_order.py::test_parse_order_too_many_colons_is_invalid PASSED                [ 40%]
test_photo_print_order.py::test_parse_order_negative_copies_is_invalid PASSED                [ 42%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[:5] PASSED                      [ 43%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[  :5] PASSED                    [ 44%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:] PASSED                    [ 46%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:abc] PASSED                 [ 47%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:2.5] PASSED                 [ 49%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[4x6:-1] PASSED                  [ 50%]
test_photo_print_order.py::test_parse_order_known_bad_inputs[] PASSED                        [ 52%]
test_photo_print_order.py::test_file_size_average_valid_lists PASSED                         [ 53%]
test_photo_print_order.py::test_file_size_average_ignores_spaces PASSED                      [ 55%]
test_photo_print_order.py::test_file_size_average_negative_size_is_invalid PASSED            [ 56%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[] PASSED                  [ 57%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[,] PASSED                 [ 59%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[4,,8] PASSED              [ 60%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[4,8,] PASSED              [ 62%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[,4] PASSED                [ 63%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[a,b] PASSED               [ 65%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[4;8] PASSED               [ 66%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[1.5,2] PASSED             [ 68%]
test_photo_print_order.py::test_file_size_average_known_bad_inputs[   ] PASSED               [ 69%]
test_photo_print_order.py::test_file_size_average_example_from_prompt PASSED                 [ 71%]
test_photo_print_order.py::test_payable_amount_valid_values PASSED                           [ 72%]
test_photo_print_order.py::test_payable_amount_example_from_prompt PASSED                    [ 73%]
```

```
test_photo_print_order.py::test_payable_amount_example_from_prompt PASSED                [ 75%]
test_photo_print_order.py::test_payable_amount_discount_larger_than_total_is_free PASSED  [ 75%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[] PASSED                  [ 76%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[450] PASSED               [ 78%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[450|] PASSED              [ 79%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[|50] PASSED               [ 81%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[|] PASSED                 [ 82%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[a|b] PASSED               [ 84%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[1|2|3] PASSED             [ 85%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[nan|1] PASSED             [ 86%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[1|nan] PASSED             [ 88%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[inf|1] PASSED             [ 89%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[1|inf] PASSED             [ 91%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[-inf|1] PASSED            [ 92%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[1e999|0] PASSED           [ 94%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[-5|1] PASSED              [ 95%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[5|-1] PASSED              [ 97%]
test_photo_print_order.py::test_payable_amount_known_bad_inputs[1.7e308|0] PASSED         [ 98%]
test_photo_print_order.py::test_payable_amount_needs_exactly_one_pipe PASSED              [100%]

====================================== 69 passed in 1.52s ======================================
```